Endocrine disruptors and hypospadias: role of genistein and the fungicide vinclozolin.

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OBJECTIVES: The phytoestrogen (plant estrogen) genistein, present in soy products, is of interest because in utero exposure to genistein can cause hypospadias in our mouse model and maternal consumption of soy is prevalent in human populations. Another compound of interest is the fungicide vinclozolin, which also causes hypospadias in the mouse and rat and can occur concurrently with genistein in the diet as a residue on exposed foods. A study in the United Kingdom found no relationship between a maternal organic vegetarian diet and hypospadias frequency, but women who consumed nonorganic vegetarian diets had a greater percentage of sons with hypospadias. Because nonorganic diets can include residues of pesticides such as vinclozolin, we sought to assess the interaction of realistic daily exposures to genistein and vinclozolin and their effects on the incidence of hypospadias. METHODS: Pregnant mice were fed a soy-free diet and orally gavaged from gestational days 13 to 17 with 0.17 mg/kg/day of genistein, 10 mg/kg/day of vinclozolin, or genistein and vinclozolin together at the same doses, all in 100 microL of corn oil. The controls received the corn oil vehicle. The male fetuses were examined at gestational day 19 for hypospadias, both macroscopically and histologically. RESULTS: We identified no hypospadias in the corn oil group. The incidence of hypospadias was 25% with genistein alone, 42% with vinclozolin alone, and 41% with genistein and vinclozolin together. CONCLUSIONS: These findings support the idea that exposure to these compounds during gestation could contribute to the development of hypospadias.